

ANKARA ESENBGA, Turkey

WMO: 171280

Lat: 40.128N

Lon: 32.995E

Elev: 953

StdP: 90.39

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.2	-10.1	-16.0	1.0	-11.8	-13.4	1.3	-7.9	10.0	3.1	8.6	2.7	1.1	30	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.0	33.9	17.0	32.1	16.7	30.6	16.5	19.1	29.1	18.1	28.3	17.4	27.4	3.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	12.2	22.9	14.3	11.4	22.2	13.6	10.9	21.5	58.4	28.8	55.1	28.2	52.6	27.4	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.5	6.6	-16.9	36.5	4.0	1.6	-19.7	37.7	-22.0	38.6	-24.3	39.5	-27.1	40.6		
(5)			-17.2	20.1	3.9	1.6	-20.0	21.2	-22.3	22.1	-24.5	23.0	-27.3	24.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-0.5	1.4	5.1	9.9	14.6	18.5	22.1	22.4	17.6	12.0	5.8	1.3		
	DBStd	8.75	4.79	4.67	4.02	3.76	3.17	2.83	2.69	2.69	3.43	3.65	3.87	4.10		
	HDD10.0	1200	326	242	158	46	3	0	0	0	0	21	133	270		
	HDD18.3	3037	584	475	411	255	122	31	3	3	52	196	377	528		
	CDD10.0	1531	0	1	5	42	145	256	377	386	229	84	6	1		
	CDD18.3	326	0	0	0	1	4	37	121	131	32	1	0	0		
	CDH23.3	4506	0	0	1	21	151	588	1569	1577	539	61	0	0		
	CDH26.7	1734	0	0	0	1	23	172	677	677	173	10	0	0		
(14) Wind	WSAvg	2.6	2.1	2.4	2.9	2.7	2.6	2.9	3.3	3.3	2.6	2.2	1.9	2.0		
(15) Precipitation	PrecAvg	421	50	39	38	46	54	35	15	10	18	27	34	57		
	PrecMax	563	104	89	93	99	111	94	64	30	55	79	113	133		
	PrecMin	266	4	8	10	6	18	9	0	0	0	0	3	4		
	PrecStd	64	30	22	20	26	23	21	14	10	17	21	27	28		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	15.9	20.9	26.0	29.0	32.3	36.2	35.8	33.1	28.1	20.1	13.1		
		MCWB	6.9	8.0	9.7	12.7	15.1	16.7	17.5	17.0	15.4	14.4	10.8	8.3		
	2%	DB	9.2	12.9	17.6	23.0	27.0	30.2	34.2	33.9	30.8	24.9	17.9	10.8		
		MCWB	5.6	6.8	8.7	11.7	14.7	16.3	17.3	17.0	15.2	13.2	10.4	6.9		
	5%	DB	7.6	10.9	15.2	20.8	25.0	28.8	32.2	32.1	28.8	22.7	15.4	9.0		
		MCWB	4.8	6.0	7.7	11.1	14.0	16.0	17.1	16.6	14.8	12.5	9.0	6.1		
	10%	DB	5.8	8.8	13.0	18.2	23.0	26.9	30.4	30.7	26.6	20.2	13.1	7.2		
		MCWB	3.9	5.1	6.8	9.9	13.5	15.4	16.7	16.6	14.4	11.8	8.0	5.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	9.1	10.7	14.2	17.0	19.2	20.6	20.4	17.8	15.6	12.3	8.9		
		MCDB	11.3	13.7	18.5	22.7	25.0	27.7	31.3	31.3	27.9	24.3	18.0	12.1		
	2%	WB	6.1	7.5	9.3	12.6	15.7	17.7	19.2	18.9	16.8	14.1	10.8	7.6		
		MCDB	8.8	11.8	15.9	21.0	24.2	26.9	29.8	28.8	26.4	21.5	16.3	10.0		
	5%	WB	4.9	6.3	8.2	11.5	14.8	16.7	18.3	18.1	15.8	13.2	9.5	6.4		
		MCDB	6.9	10.1	14.2	19.1	23.1	25.9	28.8	28.2	25.5	20.3	14.3	8.6		
	10%	WB	3.8	5.2	7.2	10.5	13.9	16.1	17.5	17.3	15.0	12.5	8.5	5.3		
		MCDB	5.6	8.3	12.4	17.2	21.6	25.0	27.9	27.5	24.9	19.0	12.6	7.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	10.1	11.8	13.0	13.8	14.6	15.4	15.0	15.3	13.7	12.7	8.3		
		MCDBR	10.7	13.8	16.5	17.4	17.2	17.5	17.8	17.1	18.5	18.1	16.8	11.0		
	5% WB	MCWBR	7.5	8.7	9.1	8.4	7.6	6.7	6.3	5.9	7.8	9.0	10.1	7.6		
		MCWBR	9.5	12.6	14.4	15.2	15.4	15.6	16.2	15.2	15.7	14.8	14.5	9.3		
(40) Clear-Sky Solar Irradiance	taub		0.302	0.328	0.372	0.409	0.387	0.368	0.365	0.364	0.344	0.344	0.313	0.303		
		taud	2.373	2.323	2.236	2.163	2.292	2.369	2.376	2.360	2.414	2.406	2.460	2.436		
	Ebn at Noon		859	883	876	864	891	906	905	898	897	855	842	830		
		Edh at Noon	85	103	126	145	130	121	119	118	104	94	78	74		
(44) All-Sky Solar Radiation	RadAvg		1.86	2.86	4.00	5.12	6.09	6.94	7.40	6.59	5.17	3.52	2.43	1.64		
	RadStd		0.23	0.43	0.37	0.48	0.47	0.60	0.32	0.30	0.37	0.26	0.24	0.22		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (9 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page